

Automatic Street Light Using IR Sensor

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Abstract—Street lighting is an important part of modern infrastructure because it provides visibility and improves safety during low-light conditions. However, continuously operating street lights can result in unnecessary energy consumption when roads are not being used. This paper presents an automatic street-light system using an Infrared (IR) sensor for detecting the presence of an object such as a vehicle or pedestrian. The detected signal is used by a control circuit to operate an LED street light automatically. The proposed system reduces the need for manual switching and provides lighting according to detected movement. The design is simple, economical, and suitable for educational prototypes and smart-lighting applications.

Index Terms—Automatic street light, IR sensor, LED, motion detection, energy efficiency, smart lighting.

I. INTRODUCTION

Street lighting plays an important role in transportation, public safety, and visibility during night-time conditions. Conventional street-lighting systems may operate for long periods even when there is little or no traffic. This can lead to unnecessary electricity consumption.

Automation can be used to control lighting according to actual requirements. Previous research has investigated the use of infrared sensors to detect vehicles and control LED street-light intensity [1]. Other systems have combined Arduino, IR sensors, and light-dependent resistors to provide automatic control according to vehicle movement and ambient light conditions [2].

In the proposed system, an IR sensor is used as the primary input for detecting movement. When an object enters the sensing range, the control circuit activates the corresponding LED street light. When the object is no longer detected, the light can be switched to its specified inactive or reduced-light state.

The main purpose of the system is to demonstrate a simple sensor-based method for automatic street-light control.

II. PROBLEM STATEMENT

Traditional street lights may remain continuously ON even when there are no vehicles or pedestrians in a particular area. Manual operation also requires human intervention.

Therefore, a system is required that can:

- Detect movement automatically.
- Control street lights without continuous manual operation.
- Reduce unnecessary lighting.

- Provide a simple and economical implementation.

III. OBJECTIVES

The main objectives of the proposed system are:

- 1) To design an automatic street-light control system.
- 2) To detect objects using an IR sensor.
- 3) To automatically control an LED street light.
- 4) To reduce unnecessary operation of the light.
- 5) To develop a simple and low-cost prototype.

IV. LITERATURE REVIEW

Ramli et al. developed an energy-efficient street-light system in which infrared sensors were used to detect vehicles and control LED light intensity. Their reported prototype compared conventional and adaptive operation and investigated energy reduction [1].

Ghosh et al. proposed a Smart Street Lighting Controller using Arduino UNO, LDR, and IR sensors. Their system used IR sensors for vehicle movement detection and LDR sensing for day/night conditions [2].

Balaji et al. investigated automatic street-light illumination control and incorporated IR sensing to improve the operation of the lighting system [3].

Singh et al. discussed automation in street lights using IR sensors, LDRs, LEDs, and microcontroller-based control techniques. Their work focused on motion detection and energy conservation [4].

Auto-intensity regulation has also been studied using Arduino, LDR, IR sensors, and real-time control techniques [5].

These studies demonstrate the usefulness of sensor-based automation for street-light control. The present work focuses on a simplified IR-sensor-based prototype suitable for educational implementation.

V. PROPOSED SYSTEM

The proposed system consists of an IR sensor, control circuit, LED street light, and power supply.

The IR sensor continuously monitors its detection area. When an object is detected, the sensor generates an output signal. The control circuit receives this signal and operates the LED street light.

The basic system architecture is shown in Fig. 1.

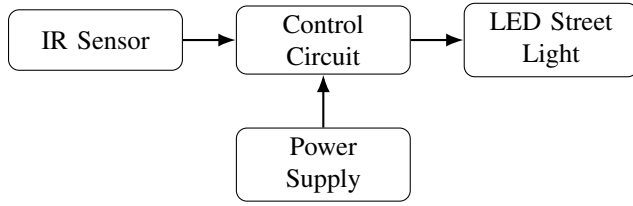


Fig. 1. Block diagram of the proposed automatic street-light system

VI. COMPONENTS REQUIRED

The main components used in the proposed prototype are listed in Table I.

TABLE I
COMPONENTS USED IN THE PROTOTYPE

Component	Purpose
IR Sensor	Detects an object or movement within its sensing range.
LED	Represents the street-light source.
Resistor	Limits current through the LED.
Control Circuit	Processes the sensor signal and controls the light.
Breadboard	Used for prototype circuit connections.
Connecting Wires	Provide electrical connections between components.
Power Supply	Provides power to the circuit.

VII. IR SENSOR

An infrared sensor can be used for proximity or object detection. A typical IR obstacle sensor contains an infrared transmitter and receiver. The transmitter emits infrared radiation and the receiver detects reflected radiation from an object.

The sensor output changes when an object enters the detection range. This output is then used by the control circuit to operate the street-light LED.

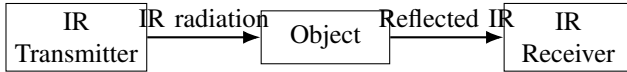


Fig. 2. Basic working concept of an IR sensor

VIII. WORKING PRINCIPLE

The working of the proposed system can be explained in the following steps:

- 1) The power supply provides power to the circuit.
- 2) The IR sensor continuously monitors the detection area.
- 3) When an object enters the sensor range, infrared radiation is reflected toward the receiver.
- 4) The IR sensor changes its output signal.
- 5) The control circuit receives the sensor signal.
- 6) The LED street light is switched ON according to the control logic.
- 7) When the object leaves the sensing area, the sensor output changes again.
- 8) The street light returns to its programmed inactive or reduced-light state.

IX. ALGORITHM

The algorithm of the proposed system is:

- 1) Start the system.
- 2) Initialize the IR sensor and LED.
- 3) Continuously read the IR sensor.
- 4) Check whether an object is detected.
- 5) If an object is detected, turn ON the LED street light.
- 6) If no object is detected, turn OFF the LED or keep it in the predefined low-light state.
- 7) Repeat the process continuously.

X. FLOWCHART

The operating sequence is represented in Fig. 3.

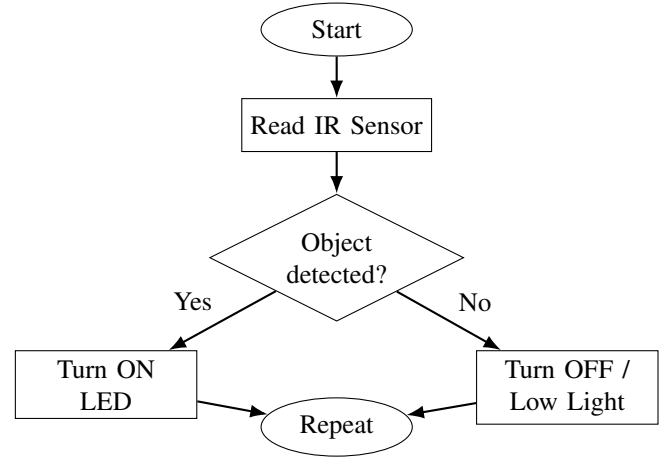


Fig. 3. Flowchart of the proposed system

XI. EXPERIMENTAL SETUP

A prototype can be assembled using an IR sensor, LED, resistor, breadboard, connecting wires, control circuit, and power supply.

The IR sensor should be positioned so that the expected object can enter its sensing range. The LED represents the street-light unit.

For a larger prototype, multiple IR sensors and LED units can be arranged along different sections of a model road.

XII. TEST CASES

The following table describes the functional test cases that should be verified on the physical prototype.

TABLE II
FUNCTIONAL TEST CASES

Test	Condition	Sensor	Expected LED
T1	No object	No detection	OFF / Low
T2	Object detected	Detection	ON
T3	Object leaves	No detection	OFF / Low
T4	Repeated movement	Detection changes	Corresponding control

Note: The table gives expected functional behavior. Actual experimental observations should be entered after testing the physical prototype.

XIII. RESULTS AND DISCUSSION

The proposed system demonstrates automatic control of an LED using an IR sensor. When an object is detected within the sensor range, the control circuit can activate the corresponding light. When the object is no longer detected, the light returns to its predefined inactive or reduced-light condition.

The experimental prototype should be evaluated by checking sensor detection, LED response, detection range, and reliability under different object positions.

Previous research has reported energy-saving benefits from adaptive street-light operation. For example, Ramli et al. reported an approximately 80% reduction in energy consumption for their specific adaptive prototype when compared with their conventional method [1]. This value should not be directly treated as the energy saving of the present prototype because the hardware and operating conditions are different.

XIV. ADVANTAGES

The proposed system provides the following advantages:

- Automatic operation.
- Reduced manual intervention.
- Simple circuit design.
- Low-cost prototype implementation.
- Movement-based lighting control.
- Suitable for educational and smart-lighting prototypes.

XV. LIMITATIONS

The system has some limitations:

- IR sensors have a limited sensing range.
- Detection depends on sensor positioning.
- Environmental conditions may affect sensor performance.
- A basic IR sensor may not be suitable for long-distance road monitoring.
- The prototype requires an appropriate power supply.

XVI. APPLICATIONS

The proposed concept can be applied to:

- Model roads and educational prototypes.
- Parking areas.
- College and institutional campuses.
- Residential roads.
- Industrial areas.
- Smart-lighting demonstration systems.

XVII. FUTURE SCOPE

The proposed system can be extended by combining IR sensors with additional sensing and communication technologies.

An LDR can be added to distinguish day and night conditions, as demonstrated in several smart street-lighting studies [2], [5]. Multiple sensors can also be used to control different sections of a road independently.

Future versions may incorporate IoT-based monitoring, solar power, wireless communication, adaptive brightness control, and fault monitoring.

XVIII. CONCLUSION

This paper presented an automatic street-light system using an infrared sensor for movement detection. The system detects an object within the sensing area and controls an LED street light according to the programmed control logic.

The proposed approach provides a simple method for demonstrating automatic lighting control and can reduce unnecessary operation of street lights. The design is suitable for an academic prototype and can be extended using LDR, IoT, solar power, multiple sensors, and adaptive intensity control.

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